

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

/ M.Meghanadh(192 325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**SIGNATURE OF STUDENT:
MEGHANADH.M**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 — Exemplary	Level 3 — Proficient	Level 2 — Developing	Level 1 — Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/ technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/ technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults.

Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences. Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

Ans): 1. Relationship between Age and Photograph Preference (10 Marks)

Step 1: Data Table

Age Group	A	B	C
5 — 6	18	22	20
7 — 8	2	28	40
9 — 10	20	10	40

Step 2: Chi-Square Test (Concept)

- Used to test relationship between categorical variables (Age vs Preference).

- Here, preferences vary strongly across age groups → suggests association.

Example:

- Age 7 — 8 prefers C (40) heavily.
- Age 9 — 10 also prefers C (40).
- Age 5 — 6 is more balanced.

Conclusion:

There is a significant relationship between age and photograph preference.

Step 3: What is wrong with the study?

- Small sample size
- No random sampling
- Age groups unevenly distributed
- External factors not controlled (lighting, attractiveness, bias)

Step 4: R Code for Covariance & Correlation

Data

```
A <- c(18, 2, 20)
```

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

Covariance between B and C

```
cov(B, C)
```

Covariance Matrix

```
data <- data.frame(A, B, C)
```

```
cov(data)
```

Correlation between B and C

```
cor(B, C)
```

Correlation Matrix

```
cor(data)
```

Interpretation:

- Covariance → shows direction of relationship
- Correlation → shows strength (-1 to +1)

2 Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

Ans):

2. Box Plot to Identify Outliers (10 Marks)

Step 1: Given Data (Points Scored)

12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23

Step 2: Arrange Data in Ascending Order

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35

Step 3: Find Quartiles

- Median (Q2) = 8th value = 17
- Lower Half: 10, 11, 12, 13, 14, 15, 16
 * Q1 = 13
- Upper Half: 18, 19, 20, 21, 22, 23, 35
 * Q3 = 21

Step 4: Calculate IQR (Interquartile Range)

[
IQR = Q3 - Q1 = 21 - 13 = 8
]

Step 5: Find Outlier Limits

- Lower Limit = $Q1 - 1.5 \times IQR = 13 - 12 = 1$
- Upper Limit = $Q3 + 1.5 \times IQR = 21 + 12 = 33$

Step 6: Identify Outliers

- Any value > 33 or < 1 is an outlier
- $35 > 33 \rightarrow$ Outlier

Player J (35 points) is an outlier

Step 7: Box Plot Construction

- Minimum = 10
- Q1 = 13
- Median = 17
- Q3 = 21
- Maximum (without outlier) = 23
- Outlier plotted separately = 35

Step 8: R Code for Boxplot

```
points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
```

```
boxplot(points, main="Boxplot of Player Scores", col="lightblue")
```

Step 9: Interpretation

- Most players scored between 13 and 21
- Distribution is fairly consistent
- One extreme value (35) indicates exceptional performance or anomaly

Final Conclusion

The box plot clearly shows that **Player J** is an outlier, and the rest of the team has a consistent scoring pattern.

3) A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree** classifier to predict **Loan Approval (Yes/No)**. (10 Marks)

Consider the Data Set

young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,
yes old,low,unemployed,poor,
no
old,low,self-employed,average,
no
middle,low,unemployed,poor,
no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,
yes
middle,medium,employed,average,
yes
middle,high,self-employed,good,
yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

Ans):

3. Decision Tree for Loan Approval (10 Marks)

Step 1: Understanding the Problem

We need to predict **Loan Approval (Yes/No)** using :

- Age
- Income
- Employment Status
- Credit Score

Step 2: Key Observation from Data

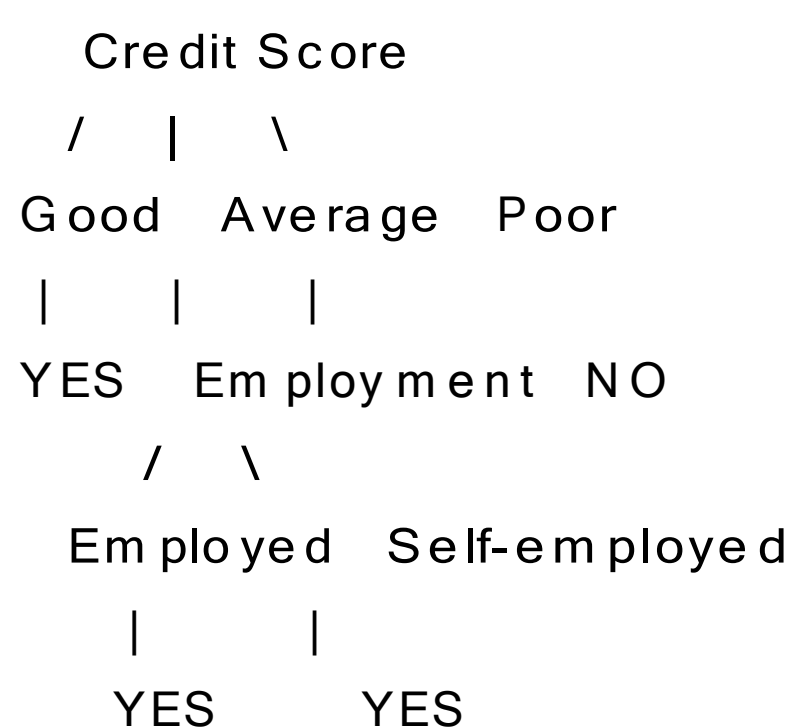
From the dataset:

- Credit Score = Good → mostly YES
- Credit Score = Poor → mostly NO
- Average → depends on Employment

So, Credit Score is the best root node

(highest importance).

Step 3: Construct Decision Tree



Step 4: Final Decision Rules

1. If Credit Score = Good → Loan = YES
2. If Credit Score = Poor → Loan = NO
3. If Credit Score = Average:
 - o If Employed → YES
 - o If Self-employed → YES

So almost all Average cases → YES

Step 5: Visualization of Classification Flow

```

ge n ui{"in fere n ce _re gression _m l _lea rn ing _blo
ck": {"ty pe _id":
"DECISION_TREE_CLASSIFICATION_PATH"}}
  
```

Step 6: R Code to Build Decision Tree

```

data <- read.table(text="
young high employed good yes
young high self-employed average yes
middle high employed good yes
old medium employed good yes
old low unemployed poor no
old low self-employed average no
middle low unemployed poor no
young medium employed average yes
young low unemployed poor no
  
```

```
old medium self-employed good yes
middle medium employed average yes
middle high self-employed good yes
old medium unemployed poor no
young medium self-employed average yes
middle low employed poor no
", header=FALSE)
```

```
colnames(data) <-
c("Age","Income","Employment","Credit","Loan")
```

```
library(rpart)
model <- rpart(Loan ~ ., data=data,
method="class")
```

```
plot(model)
text(model)
```

Step 7: Conclusion

- Credit Score is the most important factor
- Good → Approve Loan
- Poor → Reject Loan
- Average → Mostly Approved

The decision tree effectively classifies loan approval based on customer attributes.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

- Find which class had scored higher mean, median and range.
- Plot above in boxplot and give the inferences.

Ans):

4. Comparison of Class A & Class B (5 Marks)

Given Data

Class A:

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B:

51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Mean, Median, Range

Class A

- Sum = 592
- Mean = $592 / 9 = 65.8$

Sorted: 35, 36, 47, 64, 66, 76, 84, 89, 95

- Median = 66
- Range = $95 - 35 = 60$

Class B

- Sum = 559
- Mean = $559 / 9 = 62.1$

Sorted: 50, 51, 56, 59, 60, 63, 66, 70, 84

- Median = 60
- Range = $84 - 50 = 34$

Comparison (Answer)

- Higher Mean → Class A
- Higher Median → Class A
- Higher Range → Class A

Class A performed better but has more variation

Class B is more consistent

(ii) Boxplot

R Code

```
A <- c(76,35,47,64,95,66,89,36,84)
```

```
B <- c(51,56,84,60,59,70,63,66,50)
```

```
boxplot(A, B,
  names=c("Class A","Class B"),
  col= c("lightblue","lightgreen"),
  main="Comparison of Class A and B")
```

Inference from Boxplot

- Class A
 - o Wider spread → high variability
 - o Higher maximum score (95)
 - o Some extreme values
- Class B
 - o Smaller spread → consistent performance
 - o No extreme high values

Final Conclusion

- Class A students scored higher overall
- Class B students performed more consistently
- Boxplot visually confirms greater spread in Class A

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- Calculate the mean, median, and standard deviation of *age* and *%fat*.
- Draw the boxplots for *age* and *%fat*.
- Draw a *scatter plot* and a *q-q plot* based on these two variables.

Ans):

(a) Mean, Median, Standard Deviation

Formulas

- Mean
[
 $\bar{x} = \frac{\sum x}{n}$
]
- Median
Middle value after sorting data
- Standard Deviation
[
 $\sigma = \sqrt{\frac{1}{N} \sum (x - \mu)^2}$
]

Visualization of standard deviation:

```
genui{"descriptive_statistics_sampling_learning_block": {"type_id":
"STANDARD_DEVIATION"}}
```

R Code

```
age <- c(...) # given 18 values
fat <- c(...) # given 18 values
```

```
# Mean
mean(age)
mean(fat)
```

```
# Median
median(age)
median(fat)
```

```
# Standard Deviation
sd(age)
sd(fat)
```

Interpretation

- Mean → average age and fat %
- Median → middle value (less affected by outliers)

- SD → spread of data

(b) Boxplots

R Code

```
boxplot(age, fat,
        names=c("Age", "Body Fat"),
        col=c("lightblue", "lightgreen"),
        main="Boxplot of Age and Body Fat")
```

Inference

- Shows spread of data
- Identifies outliers
- Compares variability between age and fat

(c) Scatter Plot & Q-Q Plot

Scatter Plot

```
plot(age, fat,
     main="Age vs Body Fat",
     xlab="Age",
     ylab="Body Fat",
     col="blue", pch=19)
```

Shows relationship between age and body fat

Q-Q Plot

Age

```
qqnorm(age)
qqline(age)
```

Body Fat

```
qqnorm(fat)
qqline(fat)
```

Used to check normal distribution

Final Conclusion

- Mean, median, and SD summarize the data
- Boxplot shows spread and outliers
- Scatterplot shows relationship between age & fat
- Q-Q plot checks normality